GSFC NEON Series-1 CMO

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NS1-SMBA-SOW-0002, Revision C
Near Earth Orbit Network Series-1 Project, Code 473

Near Earth Orbit Network Series-1 Project Sounder for Microwave-Based Applications Statement of Work

NEAR EARTH ORBIT NETWORK
NEON
NOAA/NASA

Goddard Space Flight Center
Greenbelt, Maryland

Near Earth Orbit Network Project Series-1 Sounder for Microwave-Based Applications Statement of Work Review/Signature/Approval Page

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Preface

This document is under Near Earth Orbit Network (NEON) Series-1 (NS1) Project configuration control. Once this document is approved, NS1 approved changes are handled in accordance with Class I and Class II change control requirements as described in the Low Earth Orbit (LEO) Programs Division (LPD) Programs and Projects Configuration Management Procedure (CMP), and changes to this document shall be made by complete revision.

Any questions should be addressed to:

LEO/NEON Configuration Management Office
NASA/GSFC
Code 473
Greenbelt, MD 20771

Change History Log

Revision	Effective Date	Description of Changes
Revision –	December 20, 2024	This version incorporates 470-CCR-24-0405 which was approved by the SMBA ERB on March 26, 2024, and incorporates NS1-CCR-0003 by the NS1 SMBA CCB on the effective date shown.
Revision A	April 17, 2025	This version incorporates NS1-CCR-0011 which was approved by the NS1 SMBA CCB on the effective date shown. Editorial change all instances of climate to weather.
Revision B	August 19, 2025	This version incorporates NS1-CCR-0013 which was approved by the NS1 SMBA CCB on the effective date shown.
Revision C	January 15, 2026	This version incorporates NS1-CCR-0016 which was approved by the NS1 SMBA CCB on the effective date shown.

Table of TBDs/TBRs/TBSs

Item No.	Location	Summary	Individual/ Organization	Due Date
1	VIII.	Document numbers - The Contractor **shall** comply with all requirements in the PRD, NS1-SMBA-REQ-0001; CDRL, NS1-SMBA-CDRL-0001; ICD (TBD); and Mechanical Interface Control Document (MICD) (TBD).	Project	
2	3.3.2	The Contractor **shall** implement the launch range safety requirements that are applicable to the launch site (TBD).	Instrument Provider	
3	6.1	PHS&T Plans (included in CDRL IT-4) **shall** include one year of storage at the Contractor's facility and long term (12 years) storage plans, and necessary storage related testing, cleaning, and maintenance for FM1 and additional FMs as required. This is for planning purposes only. See contract clause (TBD).	Project	

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I. INTRODUCTION

The Sounder for Microwave-Based Applications (SMBA) is the backbone passive microwave instrument that produces global atmospheric temperature and moisture profiles for the National Oceanic and Atmospheric Administration (NOAA) and the science community. SMBA provides microwave data to support temperature and humidity soundings in cloud-covered and clear-sky conditions.

II. SCOPE

This Statement of Work (SOW) reflects the Contractor's scope within the period of performance as defined in the Contract. Throughout the remainder of the document, the Instrument supplier will be referred to as the "Contractor." The National Aeronautics and Space Administration (NASA), Goddard Space Flight Center (GSFC), and NOAA, including their support services contractors, will be referred to as the Government. The Contract, this SOW, and the Contract Data Requirements List (CDRL) (NS1-SMBA-CDRL-0001) identify the deliverables associated with the SMBA Instrument.

This SOW defines the effort required of the Contractor to design, develop, fabricate, integrate, test, analyze, verify, deliver, and provide post-launch support through commissioning for the SMBA Instrument.

This SOW covers all SMBA Instrument builds. Any requirement that is only applicable to the first flight unit or any other unit is specifically called out. Where no clarification regarding applicability is provided, the requirements are applicable to all flight units.

This SOW includes an identifier where each section or requirement applies to Development [DEV], Production [PROD], or Task Order [TO] Contract Line Item Numbers (CLIN) as per the Contract Clause GSFC 52.211-91, SCOPE OF WORK. If there is no callout, the section or requirement applies to all CLINs.

III. NEON PROGRAM AND NEON SERIES-1 PROJECT OVERVIEW

Over 2016-2018, the National Environmental Satellite, Data, and Information Service (NESDIS) initiated the National Oceanic and Atmospheric Administration (NOAA) Satellite Observing System Architecture (NSOSA) study to determine the most cost-effective space architectures for NOAA's future weather environmental remote sensing missions. Among the results of the NSOSA study was a confirmation for the continuing and evolving need for global environmental observations from low earth orbit (LEO). In response, NESDIS defined the Near Earth Orbit Network (NEON) program to meet existing and emerging global observational needs from LEO. Organizationally, the NEON program and its projects reside within the NESDIS Office of LEO Observations and the National Aeronautics and Space Administration (NASA) Goddard Space Flight Center (GSFC) Low Earth Orbit (LEO) Programs Division (LPD). The NEON Series-1 (NS1) Project is within the NEON Program.

The NS1 Project will field the next generation LEO microwave (MW) sounding observational capabilities that support Department of Commerce (DOC) Primary Mission Essential Functions (PMEF), and NOAA strategic objectives associated with terrestrial weather prediction and warning, healthy oceans, and resilient coastal communities and economies. NS1 is the follow-on microwave instrument continuity mission to the Joint Polar Satellite System (JPSS) Advanced Technology Microwave Sounder (ATMS) observations. Satellite-based Microwave temperature and humidity soundings to be provided by the NS1 satellites are the two largest impact sources of observations for forecasts. NS1 will utilize microwave sounding to provide necessary atmospheric state information for numerical weather prediction and other applications. The National Weather Service's weather products need three-dimensional atmospheric state information which is provided by atmospheric sounding in the form of vertical profiles (state) of atmospheric water vapor and temperature.

The NS1 Project scope includes satellites that host low earth orbit (LEO) Earth-observing microwave sounding instruments, related ground services (i.e., command and control, and data routing, processing, distribution, and archival), and launch services. The NS1 mission consists of flight, launch, and ground segments along with the interfaces necessary for the collection of environmental data. This includes the periodic data downlink to ground stations and continuous Direct Broadcast (DB) allowing for data product generation and distribution to users. NS1 consists of single instrument satellites operating in a sun synchronous orbit each with a design life of 5-years. It includes development, integration, test, launch, and science operations of the Sounder for Microwave-Based Applications (SMBA) instrument. The SMBA instrument will produce global atmospheric temperature and moisture profiles allowing the NEON Program to provide science continuity with ATMS while introducing the addition of more frequency bands, hyperspectral data collection, and Radio Frequency Interference (RFI) detection capabilities.

NS1 will utilize best practices and processes of the commercial space industry to meet Federal program level requirements while delivering these critical SMBA hyperspectral microwave sounding observation capabilities utilizing small satellites that can be flexibly deployed.

IV. DEFINITIONS

Contractor – The developer of the item of reference. If the term Contractor is used in this document without specific reference to an item (e.g., SMBA, ground support equipment (GSE), etc.), then the term **shall** be interpreted to imply the SMBA Contractor. In all cases, the term Contractor also implies any and all associated suppliers and subcontractors.

Critical – Subsystems of higher risk/complexity, significant dollar/schedule value, or near/on the critical path as identified in the Engineering Peer Review (EPR) plan (CDRL PM-2).

Government – Federal Civil Servants and support contractors (contractors that represent NASA or are delegates for NASA).

Shall – Compliance by the Contractor is mandatory. Any deviations from these contractually imposed mandatory requirements require the approval of the Contracting Officer.

Should – Preferred good practices are identified by "should" statements. "Should" designates performance, which is requested, but not mandatory. Noncompliance of a "should" statement does not require Government approval but should be addressed at the earliest applicable project milestone.

May – At the discretion of the Contractor or Government.

Will – Designates the intent of the Government.

Applicable documents – The term "applicable documents" denotes Government-prepared and controlled documents and industry standards documents that levy additional requirements on the Contractor. The Contractor is required to comply with the documents, or the tailored version of the documents provided by the Government.

Reference documents – The term "reference documents" denotes documents that amplify or clarify the information presented in this document. The Contractor is not required to meet any requirements listed within these documents.

TBS – The term "to be supplied" (TBS) refers to a document or information that will be developed or provided.

TBR – The term "to be reviewed" (TBR) means that the stated requirements or values will be reviewed for appropriateness jointly by the Contractor and the Government and the information may be changed. The TBR requirement/value remains a requirement until formal process of configuration change review, approval, and Contract modification.

TBP – The term "to be proposed" (TBP) indicates required Contractor fill-ins as part of the solicitation process.

TBD – The term "to be determined" (TBD) identifies a missing requirement or value; indicates further research or analysis is needed to determine the information. All changes to resolve these items will be completed through a formal process of configuration change review, approval, and Contract modification.

Instrument – The SMBA flight hardware, support equipment (i.e., system test equipment (STE)/GSE), software, documentation, and/or data necessary to fulfill the requirements of the Contract and this SOW.

Satellite – The term "satellite" refers to the NEON Series-1 spacecraft bus integrated with the flight Instrument(s). The term "satellite" is equivalent to the term "satellite" used in other NASA and NEON Program documents.

Virtual/Virtually – The terms "Virtual" or "Virtually," when applied to products being provided, indicate that the products will be provided electronically (e.g., via GSFC SharePoint).

Class I – A change that affects a configuration item's form, fit, or function. This could include changes to baselined documentation, technical requirements, Contract end items, interfaces, weight, power, data rate allocations, technical risks, or science performance. All proposed Class I changes require Configuration Control Board (CCB) approval.

Class II – A change that does not affect a configuration item's form, fit, or function. This could include changes to documentation only, minor hardware changes, or drawing changes that do not affect a baseline or interface. Class II changes do not require Project CCB review unless they will be written against Configuration Management (CM)-controlled documents.

V. APPLICABLE DOCUMENTS

The Applicable Documents listed in this section apply directly to the performance of this SOW to the extent specified and invoked by this SOW. The Contractor **shall** comply with the documents to the extent specified herein. The Contract, Attachment N, provides the applicable revision and issue date as well as any additional Configuration Change Requests (CCR) for the applicable documents.

Document Number	Document Name
NS1-SMBA-CDRL-0001	SMBA Contract Data Requirements List (CDRL)
NS1-SMBA-REQ-0001	SMBA Performance Requirements Document (PRD)
AS9100	Quality Management Systems
AS9003	Inspection and Test Quality Systems Requirements for Aviation, Space, and Defense Organizations
NASA-STD-8739.4	Workmanship Standard for Crimping, Interconnecting Cables, Harnesses, and Wiring
IPC/WHMA-A-620x-S	Requirements and Acceptance for Cable and Wire Harness Assemblies – Space Addendum
NASA-STD-8719.24 with Annex	NASA Payload Safety Requirements
GSFC-STD-1001	Criteria for Flight and Flight Support Systems Life Cycle Reviews
NASA-STD-8739.6	Implementation Requirements for NASA Workmanship Standards
IPC J-STD-001, class 3	Standard Soldering Requirements
GSFC-STD-6001	Area Array Package Assembly and Manufacturing Practices for Flight Hardware
IPC-2225	Sectional Design Standard for Organic Multichip Modules (MCM-L) and MCM-L Assemblies
ANSI/ESD S20.20	Protection of Electrical and Electronic Parts
IPC-6015	Qualification and Performance Specification for Organic Multichip Module (MCM-L) Mounting and Interconnecting Structures
IPC-6012	Qualification and performance of rigid printed boards
MIL-PRF-55110	Performance Specification: Printed Wiring Board, Rigid General Specification
ECSS-Q-ST-70-60	Qualification and procurement of printed circuit boards
GEIA-STD-0005-1	Standard for Developing a Lead-Free Control Plan to Manage the Risks of Lead-Free Solders and Finishes in Aerospace, Defense, and High-Performance Soldered Electronic Products
GEIA-STD-0005-2	Standard for Mitigating the Effects of Tin Whiskers in Aerospace and High Performance Electronic Systems
NASA-STD-8739.12	Metrology & Calibration
GSFC EEE-INST-002	Instructions for Electrical, Electronic, and Electromechanical (EEE) Parts Selection, Screening, Qualification, and Derating
NASA-STD-6016	Standard Materials and Processes Requirements for Spacecraft
NASA-STD-6030	Additive Manufacturing Requirements for Crewed Spaceflight Systems Foundational Principles

MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
NASA-STD 8719.14	Process for Limiting Orbital Debris
NPR 8715.7	Payload Safety Program
MSFC-STD-3029	GUIDELINES FOR THE SELECTION OF METALLIC MATERIALS FOR STRESS CORROSION CRACKING RESISTANCE IN SODIUM CHLORIDE ENVIRONMENTS
N/A	Integrated Program Management Data and Analysis Report (IPMDAR) Implementation & Tailoring Guide
DI-MGMT-81861	Data Item Description for Integrated Program Management Data and Analysis Report (IPMDAR)
N/A	CPD FFS 20200312 Contract Performance Dataset - FFS.pdf (osd.mil)
N/A	SPD FFS 20200312 Schedule Performance Dataset - FFS.pdf (ssitools.com)
N/A	NASA IPMDAR Tailoring Supplement (NITS)
NIST SP 800-37	Risk Management Framework for Information Systems and Organizations: A System Life Cycle Approach for Security and Privacy
NIST SP 800-53	Security and Privacy Controls for Information Systems and Organizations
NASA-STD-1006	Space System Protection Standard
NPR 1600.1	NASA Security Program Procedural Requirements
NPR 1040.1	NASA Continuity of Operations Planning Procedural Requirements Document

VI. REFERENCE DOCUMENTS

The documents listed in this section are those documents referenced in the SOW which contain information relating to the work required, but which do not constitute requirements.

Document Number	Document Name
AS9146	Foreign Object Damage (FOD) Prevention Program
NPR 8000.4	Risk Management Procedural Requirements
GSFC 500-PG-8700.2.8	FPGA Development Methodology
GSFC 500-PG-8700.2.7	Procedures and Guidelines for Design of Space Flight Field Programmable Gate Arrays
GSFC-STD-7000	General Environmental Verification Standard (GEVS)
GPR 8700.6	Engineering Peer Reviews
GPR 8700.4	Integrated Independent Reviews
ANSI/EIA748	Industry Guide for Earned Value Management
IEST-STD-CC1246	Product Cleanliness Levels and Contamination Control Program
470-LEO-PROC-00010	Low Earth Orbit (LEO) Programs Division (LPD) Configuration Management Procedure
FAR Supplement Clause 1852.234-2	NASA Federal Acquisition Regulation (FAR) Supplement Clause 1852.234-2
NASA-STD-8739.8	Software Assurance and Software Safety Standard
NASA-TM-86556	Lubrication Handbook for the Space Industry (Part A: Solid Lubricants, Part B: Liquid Lubricants)
NASA/CR-2005-213424	Lubrication for Space Applications
SP-2010-403	NASA Schedule Management Handbook
NASA SP-2016-6105	NASA Systems Engineering Handbook Appendix G

VII. PRECEDENCE

In the case of conflicting requirements, the order of precedence of documents not specifically called out in the Contract is: The Contract takes precedence over the SOW, followed by the CDRL, PRD, Interface Control Documents (ICD), and the remaining applicable documents. In the event of an inconsistency between any of the reference documents herein and the requirements of this SOW, the SOW **shall** take precedence. Any conflicts discovered by the parties between the requirements in this SOW and any referenced specification herein **shall** be brought to the attention of the other party for resolution.

VIII. WORK TO BE PERFORMED

This section, along with the Contract, CDRL (NS1-SMBA-CDRL-0001), and all documents invoked within these documents, describes the work to be accomplished by the Contractor. Specifically, the Contractor **shall** provide the effort and resources that include, but are not limited to, personnel, materials, equipment, and facilities necessary for the successful and on-time design, analysis, development, fabrication, assembly, integration, test, calibration, qualification, delivery, and post-delivery support of the Instrument. "Fabrication" in this sense also includes procured items.

The Contractor **shall** comply with all requirements in the PRD (NS1-SMBA-REQ-0001); CDRL, (NS1-SMBA-CDRL-0001); ICD (**TBD**); and Mechanical Interface Control Document (MICD) (**TBD**). The Contractor is providing input to the ICD and MICD in CDRL SE-11.

The Contractor **shall** deliver to the Government Instruments that are fully tested, qualified, and calibrated. The Contractor **shall** support the development of bilateral Instrument documents, analyses, and models, which define the details of the Instrument-to-spacecraft interface and Instrument accommodation and operating information.

The Contractor **shall** deliver numeric data in a digital format that is not a scanned format.

The Contractor **shall** provide all necessary effort to produce and deliver all materials required in the SMBA CDRL and Data Item Description (DID) (NS1-SMBA-CDRL-0001). Call-outs for individual CDRLs within this SOW are stated as a convenience. Not all required CDRL deliverables and the necessary efforts to develop those CDRLs are stated in this SOW.

1 MANAGEMENT

1.1 Program Management

The Contractor **shall** assign a point of contact to the Government Instrument Manager/Contracting Officer Representative who is responsible for the performance of the contractual effort.

1.1.1 Government Insight

The Contractor's management, tracking, control, and reporting processes **shall** facilitate insight by the GSFC Program Office into the Instrument progress and status, including Contractor systems necessary to share information and review material within the scope of the contract.

The Contractor **shall** share information with the Government via the Government's on-line collaborative workspace.

The Contractor **shall** support weekly meetings and telecons with the Government to discuss status, progress, and planned work; technical issues; financial and business metrics such as staffing, spending, and funding; and special studies' task status and hours.

The Contractor **shall** invite the Government to critical subcontractor reviews and ensure Government access to their data within the scope of the Contract.

1.1.2 Subcontract Management

The Contractor **shall** ensure that all subcontractors establish and maintain technical, cost and schedule baselines.

1.1.2.1 Flow Down of Requirements to Subcontractors

The Contractor **shall** flow all applicable requirements and controls stated or referenced herein to its Subcontractors.

1.2 Financial Management and Schedule Reporting [DEV]

The Contractor **shall** use and maintain an Earned Value Management System (EVMS) in accordance with the EVMS requirements described in this Contract Section I, Clause NFS 1852.234-2. The EVMS **shall** be supported by the Contractor's management processes and systems and in accordance with the Contractor's Earned Value Management (EVM) Plan. Reference the Contract Performance Report delivered as part of the Integrated Program Management Data and Analysis Report (IPMDAR) (CDRL PM-3).

The Contractor **shall** provide notification of all Baseline Change Requests (BCR) (CDRL PM-1). The Contractor **shall** obtain Government approval for BCRs that are over $500,000.

1.3 Risk Management

The Contractor **shall** establish and maintain a comprehensive risk management program that meets the requirements of the Contractor's established Risk Management processes or NASA Procedural Requirement (NPR) 8000.4, Risk Management Procedural Requirements.

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The Contractor **shall** invite the Government to attend Contractor Risk Management Board meetings.

1.4 Configuration Management

The Contractor **shall** implement a configuration control process that enables the Contractor Program Manager (PM) to assess and control configuration planning, identification, control, status accounting, and verification.

The Contractor **shall** submit for Government consideration a waiver or deviation for any flight item that is considered non-compliant with the requirements.

1.4.1 Configuration/Change Control

The Contractor **shall** notify the Government as soon as practical for any changes, waivers, or deviations that would affect the requirements of this contract or its execution. Impacts to the ICD or MICD (i.e., Instrument-to-spacecraft interfaces) are highly sensitive to early identification and coordination among all affected parties.

1.4.1.1 Design Changes

The Contractor **shall** document and control engineering changes and deliver to the Government (CDRL SE-4).

1.5 Reviews and Meetings

The Contractor **shall** conduct, host, prepare, and present Instrument reviews and provide review packages in accordance with the stated CDRL/DID (NS1-SMBA-CDRL-0001) for each sensor Instrument as shown in Table 1-1. The Government **shall** be invited to participate in all required reviews. All reviews and meetings **shall** provide a virtual attendance option, unless otherwise agreed to by the Government.

The top-level Instrument reviews are provided in Table 1-1 below. Further details for each review can be found in the subsequent sections of this SOW.

The Contractor **shall** develop and apply a process for capturing, responding, and closing assigned action items assigned by the Government; reviews are not complete until actions are complete or a detailed plan for closure is submitted by the Contractor and approved by the Government.

Table 1-1. Top-Level Instrument Reviews

Review	**Applicability**	**CDRL Reference**	**Chair**
Integrated Baseline Review (IBR)	[DEV]	RE-8	Government
System Requirements Review (SRR)/ System Definition Review (SDR)	[DEV]	RE-3	Government

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Review	Applicability	CDRL Reference	Chair
Preliminary Design Review (PDR)	[DEV]	RE-4	Government
Critical Design Review (CDR)	[DEV]	RE-5	Government
Manufacturing Readiness Review (MRR)	All Flight Instruments	RE-10	Contractor
Pre-Environmental Review (PER)	All Flight Instruments	RE-6	Contractor
Pre-Ship Review (PSR)	All Flight Instruments	RE-7	Government

1.5.1 Review Details

1.5.1.1 Kickoff Meeting

The Contractor **shall** hold a Kickoff Meeting [DEV] (CDRL RE-1).

The Contractor **shall** hold a Production Kickoff Meeting [PROD] for flight model (FM)-3 and beyond (CDRL RE-11). Production Kickoff Meetings will be conducted for each Option (e.g., Option 1, FM3&4). The Contractor may combine the Production Kickoff Meeting with the Manufacturing Readiness Review (MRR) if there are minimal changes between FMs.

1.5.1.2 Program Management Review

The Contractor **shall** hold monthly Program Management Reviews (CDRL PM-1).

1.5.1.3 Engineering Peer Reviews

The Contractor **shall** conduct EPRs in accordance with the EPR Plan (CDRL PM-2) and invite the Government to attend. The EPR plan identifies the critical subsystems and suppliers that are referred to in the SOW and CDRL documents whenever references are made to critical subsystems and suppliers.

The Contractor **shall** complete subsystem or lower-level peer reviews prior to any Instrument design reviews and completion of these subsystem or lower-level peer reviews are considered part of the entry criteria in the formal review process.

1.5.1.4 Manufacturing Readiness Review

The Contractor **shall** prepare for and conduct an MRR (CDRL RE-9).

1.5.1.5 Test Data Reviews

The Contractor **shall** invite the Government personnel to Quicklook reviews and Post-Test data reviews.

1.5.1.6 Test Process Decision Meetings

The Contractor **shall** conduct Test Readiness Reviews (TRR) in accordance with CDRL RE-10 prior to the start of any test of the Instrument and critical subcontractor/subsystem as identified in the EPR Plan (CDRL PM-2).

The Contractor **shall** include Government personnel in TRRs and in the test procedure review process and provide a copy of the final procedure (CDRL IT-3).

The Contractor **shall** conduct, document, and invite the Government to Consent to Proceed (CTP) and Consent to Break (CTB) (CDRL IT-3) as informal EPRs and make available upon request any data presented or referenced in support of any decision related to the forward processing of the Instrument. CTPs/CTBs **shall** be conducted at the end of: the initial Comprehensive Performance Test (CPT), Electromagnetic Interference (EMI)/Electromagnetic Compatibility (EMC), Vibration, Shock and Acoustic, Thermal Vacuum (TVAC), plateau changes in TVAC, and at final CPT. The Contractor **shall** establish criteria ahead of the meetings and evaluate whether the criteria were met.

1.5.1.7 Technical Interchange Meetings and Other Reviews and Meetings

The Contractor **shall** conduct technical/management meetings with the Government as needed or as requested to discuss and resolve any technical and/or managerial issues or concerns; any actions **shall** be managed by the Contractor to completion.

1.5.2 Government-Chaired Instrument Reviews

All Government-chaired reviews will have review boards appointed by the Government. The Contractor **shall** propose, for Government chair approval, one independent senior engineer or PM familiar with the Instrument to this review board. The Contractor **shall** determine the optimum number of consecutive days to perform each of these reviews to meet the requirements in GSFC-STD-1001, Criteria for Program Flight Critical Milestone Reviews, as identified in the associated CDRL DIDs. Formal action items are logged and tracked by the Government Review Team. The Contractor **shall** demonstrate compliance with the review success criteria identified in the associated CDRL DIDs (CDRLs RE-3, RE-4, RE-5, RE-7, RE-8). The Government will work with the Contractor to refine the review agenda.

If the review success criteria are not met to the Government's satisfaction, a “delta” review **shall** be conducted. This “delta” review **shall** meet the requirements of the appropriate system-level review (e.g., Critical Design Review (CDR)) and must cover all updates, modifications, and impacted designs and analyses as defined herein and in the applicable CDRL.

1.5.2.1 Integrated Baseline Review [DEV]

The Contractor **shall** prepare for and support an Integrated Baseline Review (IBR) to assess and validate the Performance Management Baseline in accordance with CDRL RE-7. This review will be scheduled as early as practicable and conducted in accordance with NASA requirements within 180 days of Contract award or formal Contract definitization.

1.5.2.2 System Requirements/Definition Review [DEV]

The Contractor **shall** conduct a System Requirements Review (SRR)/System Definition Review (SDR) (CDRL RE-2).

1.5.2.3 Preliminary Design Review [DEV]

The Contractor **shall** conduct a Preliminary Design Review (PDR) at the conclusion of the preliminary design effort (CDRL RE-3).

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1.5.2.4 Critical Design Review [DEV]

The Contractor **shall** conduct a CDR (CDRL RE-4).

The Contractor **shall** obtain Government concurrence to purchase any parts and materials prior to CDR.

The Contractor **shall** ensure all subsystems have achieved Technology Readiness Level (TRL) 6 maturity by CDR.

1.5.2.5 Pre-Environmental Review

The Contractor **shall** conduct a Pre-Environmental Review (PER) prior to the start of environmental testing of each FM Instrument to establish the readiness of the system to enter environmental test and to evaluate the readiness of environmental test plans and procedures.

1.5.2.6 Pre-Ship Review

The Contractor **shall** conduct a Pre-Ship Review (PSR) (CDRL RE-6) at the conclusion of the integration and test (I&T) effort prior to shipment and/or storage of each SMBA FM. If the Instrument is stored for a period greater than six months prior to shipment, the Contractor **shall** prepare an abbreviated (half-day) delta-PSR immediately prior to shipment to review logistical shipping details as well as any additional work or testing performed on the Instrument since the PSR.

1.5.3 Higher-Level Reviews

Once the Instrument is delivered, the Contractor **shall** support all reviews necessary for the successful integration, test, launch, and operation of this Instrument as defined in Section 6 of this SOW.

The Contractor **shall** support the project and satellite-level reviews (CDRL RE-11). The Contractor **shall** support these reviews by contributing to, preparing, reviewing, and/or presenting presentations and documentation; see CDRL RE-11, Inputs to Project Level Reviews. The reviews will be conducted at the spacecraft contractor's facilities or GSFC:

- Integration Readiness Reviews (1 day for each FM).
- Satellite I&T reviews (as necessary during I&T).
- Satellite Operations Reviews (e.g., end-to-end testing, operations readiness).
- Satellite Pre-environmental Review (3 days for each FM).
- Satellite Pre-storage Review (3 days for each FM).
- Satellite Pre-ship Review (3 days for each FM).
- Project SRR/SDR (3 days).
- Project PDR (3 days).
- Project CDR (3 days).
- Project Mission Operations Review (1 day per FM) [DEV, TO].
- Project Operational Readiness Review (1 day per FM) [DEV, TO].
- Project Post-Commissioning Review (1 day per FM) [DEV, TO].

1.6 Data Retention

Consistent with Contract Section E, Clause GSFC 52.246-102, INSPECTION SYSTEM RECORDS, the Contractor **shall** retain those data necessary to provide a technical and program database of significant decisions, plans, and the execution thereof as well as those data providing design, manufacturing, and test history of all deliverable items for the period of 6 years after delivery of all items and completion of all services called for by the Contract. Data to be retained include the following:

- Part and material procurement: Test, screening, and inspection data, including nonconformance reports.
- Subassembly and higher assembly: Fabrication, assembly, test, and inspection data.
- Shop travelers (including travelers at Subcontractors).
- Kitting/traceability records.
- Executed test procedures.
- Listing of test equipment used.
- Test and calibration data, including nonconformance reports.
- GSE certification, calibration of test equipment.
- As-built/as-designed verifications.
- Operating time logs.
- Total test time for each environment.
- End Item Data Packages.
- Design review data and packages.
- Drawings.
- Contractor Parts, Materials, and Processes.
- Control Authority records of all performed testing:
    - Qualification and acceptance test data.
    - Destructive physical analysis reports with samples.
    - Radiation hardness assurance test data.
- Computer Aided Design data files that capture the mechanical design of the Instrument and associated GSE.
- Photographic documentation.
- Models used to perform analyses for delivered CDRLs (e.g., thermal, optical, contamination, General Reflector Antenna Analysis Software Package).
- Environmental records.
- Numeric data in digital format.
- Software lookup tables.
- Sensor data collected during formal testing.
- Documentation of National Institute for Standards and Technology (NIST)-traceability related to calibration or intercalibration, if applicable.

1.7 Security

The Contractor **shall** maintain the Federal Information Systems in accordance with the NIST Risk Management Framework (RMF) identified in Risk Management Framework for Information Systems and Organizations: A System Life Cycle Approach for Security and Privacy, NIST SP 800-37, and Security and Privacy Controls for Information Systems and Organizations, NIST SP 800-53. Per NIST SP 800-37 (Appendix B and Appendix G), all systems that operate on behalf of NASA or that store or process data on behalf of NASA are Federal Information Systems subject to the same NIST SP 800-53 security and privacy requirements as NASA-owned systems.

The Government will follow Space System Protection Standard, NASA-STD-1006, to prepare and submit Candidate Protection Strategies (CPS) leading to a Project Protection Plan (PPP). The Contractor **shall** support these NASA-led CPS and PPP activities and updates throughout the lifecycle.

The Contractor **shall** respond to NASA-developed project-specific threat impact assessments and implement mitigations for threats, as needed.

The Contractor **shall** develop and implement a plan for ensuring security and emergency responses in accordance with NASA Security Program Procedural Requirements, NPR 1600.1, and NASA Continuity of Operations Planning Procedural Requirements, NPR 1040.1.

The Contractor **shall** implement and monitor physical security controls in all areas wherein SMBA systems are being developed or manufactured.

2 SYSTEMS ENGINEERING

The Contractor **shall** implement a systems engineering approach applied to all Instrument-related activities throughout its stages of development that includes such responsibilities as:

a) Providing technical direction and oversight throughout all phases of the program.
b) Leading, supporting, and/or preparing documentation for peer reviews as well as those reviews defined in Section 1.5.
c) Performing coordination, studies, and analyses required to interface the Instrument to the spacecraft, including Instrument specific support to ground system requirements development and operations.
d) Completing analyses and simulations in support of technical requirements compliance, verification, and demonstrations to fully establish, define, maintain, and control allocations for all required performance and design parameters.

2.1 Requirements Deviations, Waivers, Analyses, Derivations, and Allocations

The Contractor **shall** provide the definition, allocation, derivation, and traceability of requirements and the verification approach (CDRL SE-1).

The Contractor **shall** deliver verifications for NS1-SMBA-REQ-0001 PRD requirements (CDRL SE-1).

Requests for deviations or waivers to the Government's requirements **shall** be submitted to the Government's Contracting Officer for Government approval and documented as Engineering Change Requests (CDRL SE-4).

2.1.1 Technical Performance Metrics

The Contractor **shall** identify and manage Technical Performance Metrics (TPM) (CDRL SE-2), comprised of key performance requirements for the end item product, including tracking and reporting TPM status and changes (CDRL PM-1).

2.1.2 ICD and MICD

The Contractor **shall** develop and deliver input to an Instrument-to-spacecraft ICD/MICD (CDRL SE-11) that includes the Instrument's interface requirements that will be provided to the spacecraft contractor to form the basis of the Instrument-to-spacecraft ICD/MICD. The Contractor **shall** collaboratively work with the Government and the spacecraft contractor to review and maintain accurate Instrument-to-spacecraft ICD(s) and MICD(s).

The Contractor **shall** collaborate with the spacecraft contractor to implement an ethernet-type data interface between the Instrument and spacecraft.

2.1.3 Giver/Receiver

The Contractor **shall** collaboratively support the development, maintenance, and review of the SMBA giver/receiver (GR) list to be exchanged between the Contractor and other organizations. The Contractor **shall** provide the items documented in the GR list per the schedules documented and agreed to therein.

The GR list **shall** be incorporated into the IMS in accordance with CDRL PM-3, IPMDAR.

2.2 Contamination Control

The Contractor **shall** incorporate contamination control into the Instrument design, processes, and planning per IEST-STD-CC1246, PRD Section 5.7.7, and in accordance with CDRLs SE-8, IT-1, and IT-4. The Instrument **shall** be maintained at a cleanliness level as defined in the Inputs to Instrument-to-Spacecraft ICD (CDRL SE-11). The Instrument **shall** be cleaned prior to its use, shipment, and integration onto the spacecraft. Vent locations and paths **shall** be defined in the Inputs to Instrument-to-Spacecraft MICD (CDRL SE-11). The Contractor **shall** monitor the Instrument for contamination effects and validate Instrument contamination controls. The Contractor **shall** document contamination analyses as specified in the analysis reports in accordance with CDRL SE-8.

The Contractor **shall** verify that all contamination requirements are met prior to delivery to the spacecraft facility.

If the Instrument does not have a purge port accessible after delivery and integration, the Contractor **shall** conduct a contamination study to determine the best value approach to protecting the Instrument from contamination sources that exist through launch and deliver the data, results, and recommendations to the Government for disposition.

The Contractor **shall** design the Instrument and its thermal control systems to enable powered operations sufficient to conduct aliveness and limited performance testing while in a clean room.

The contractor **shall** verify that the Instrument can operate while bagged and define how long it can operate while bagged with respect to thermal constraints.

The Contractor **shall** supply any Instrument bags needed after delivery to the spacecraft.

The Contractor **shall** be responsible for Instrument cleaning and sampling post-delivery through encapsulation.

The mechanical GSE **shall** be delivered in a state that prevents particulate contamination.

All GSE intended to be used in spacecraft-level TVAC **shall** be baked out prior to delivery to the spacecraft contractor.

The contractor **shall** plan and implement a Foreign Object Damage/Debris (FOD) program that prevents the introduction of FOD into the Instrument, using AS9146, FOD Prevention Program – Requirements for Aviation, Space, and Defense Organizations, as a guideline.

2.3 Design Engineering

The Contractor **shall** design and deliver an Instrument that meets all applicable requirements included in the PRD, CDRL, ICD, and MICD.

2.4 Design, Analysis, and Verification Documentation

The Contractor **shall** perform and document all analyses of the data and information from the design, qualification testing, acceptance testing, compatibility testing, and on-orbit testing of the

Contractor's hardware and software which are required to ensure that the Instrument will meet its specifications and objectives. All analyses performed by the Contractor under this Contract **shall** be made available to the Government upon request.

2.4.1.1 Thermal Analyses and Modeling [DEV]

The Contractor **shall** develop and deliver a comprehensive thermal analysis, a Thermal Analysis Report documenting the analysis, and Thermal Models (CDRL SE-7). Thermal analyses **shall** be performed to demonstrate that test configurations and associated GSE (e.g., cold plates, chillers, chambers, thermal targets) have sufficient control authority to TVAC cycle the flight hardware to protoflight levels and simulate mission environments during thermal balance tests. The detailed analysis **shall** demonstrate that all parts meet established thermal design criteria as called out in the SMBA PRD and based upon de-rating or electrical performance requirements.

Thermal verification tests **shall** be configured so that heat pipes are in gravity neutral or reflux orientations to ensure operation of these devices during thermal testing.

2.4.1.2 Electromagnetic Compatibility/Electromagnetic Interference Analysis

The Contractor **shall** perform EMC/EMI analysis and/or test in accordance with the SMBA PRD (NS1-SMBA-REQ-0001) (CDRL SE-9).

2.4.2 Trending

The Contractor **shall** establish and implement a method for trending test data through the life of the program (CDRL SE-2).

2.5 Special Studies [TO]

The Contractor **shall** conduct, in addition to the requirements specified in this document and the Contract, additional engineering studies, tests, technical analyses, reviews of test results, design modifications, risk reduction activities, and tasks relating to the development, implementation, characterization, and operation of the Instrument, as authorized by the Government and in accordance with the Contract clauses applicable to CLIN 0003. Each task will be initiated by written direction from the Government Contracting Officer. The Government will coordinate with the Contractor to define each task in detail, and establish manpower ceilings, performance schedules, and deliverables.

3 MISSION ASSURANCE

The Contractor **shall** develop, implement, and maintain a comprehensive mission assurance program.

3.1 General

3.1.1 Management

The Contractor **shall** designate an independent Point of Contact for assurance activities that is independent of project management and with functional freedom and authority to interact with all elements of the project.

3.1.2 Surveillance

The Contractor **shall** report the status of facility operations and quality metrics to NASA on a quarterly basis. This **should** include the data for first tier and second tier suppliers in the Critical Items List.

3.2 Quality Management System

3.2.1 General

The prime Contractor **shall** have a quality management system (QMS) that meets the intent with the requirements of Society of Automotive Engineers (SAE) AS9100, "Quality Systems - Aerospace - Model for Quality Assurance in Design, Development, Production, Installation, and Servicing."

The Contractor **shall** invite the Government to the Contractor's Anomaly Review Board (ARB), Failure Review Board (FRB), Material Review Board (MRB), Parts Control Board (PCB), and Materials and Processes Control Board where contract requirement compliance may be impacted as a result of decisions made.

Subcontractors and suppliers of critical hardware systems (e.g., components, subassemblies), piece parts, or suppliers using special processes determined to be critical (e.g., plating, polishing, soldering, brazing, additive manufacturing) **shall** have a QMS that meets the intent of AS9003B.

3.2.2 Material Review Board

The contractor **shall** process major non-conformances through an MRB, which are those that affect form, fit, or function.

3.2.3 Anomaly Reporting and Disposition

The Contractor **shall** implement an ARB to process (i.e., report and disposition) major anomalies, which are those that have resulted in hardware or software test failures and damage or potential damage to hardware.

Reporting of major anomalies (CDRL PM-1) **shall** begin with the first application of power at the flight component level, at flight software acceptance testing, when interfacing with flight hardware, or at the first mechanical operation.

The contractor **shall** report accidents, test failures, or other mishaps and close calls promptly to the Government as per the Contractor's Mishap Plan.

3.3 System Safety

3.3.1 General

The Contractor **shall** document and implement a system safety program and plan in accordance with NPR 8715.7, Payload Safety Program (CDRL MA-3).

3.3.2 Mission Related Safety Requirements Documentation

The Contractor **shall** implement the launch range safety requirements that are applicable to the launch site (**TBD**).

3.3.3 Safety Requirements Compliance Checklist

The Contractor **shall** demonstrate that the payload complies with NASA and range safety requirements (CDRL MA-4).

3.3.4 Instrument Safety Assessment Report

The Contractor **shall** generate an Instrument Safety Assessment Report (ISAR) (CDRL MA-5).

The Government will provide the ISAR (CDRL MA-5) as an input to the spacecraft contractor's Satellite Safety Data Package.

3.4 Reliability

3.4.1 Reliability Program Plan

The Contractor **shall** implement a Reliability Program that includes both qualitative and quantitative techniques to support decisions regarding mission success and safety throughout system development.

3.4.2 Failure Modes, Effects, and Criticality Analyses

The Contractor **shall** perform, document, maintain, and deliver a Failure Modes, Effects, and Criticality Analysis (CDRL MA-6) in accordance with an industry Voluntary Consensus Standard (e.g., SAE) to identify and rank critical items that are being designed, built, or provided from project initiation through launch and mission operations.

The Contractor **shall** perform a Limited Life Item Analysis that includes expected life, required life, an assessment of life margin (including servicing and maintenance); provides retention rationale for items with an expected life of less than 2x the required life; and fosters a plan to manage limited life items (Life Test Plan and Final Report for Lubricated Mechanisms, CDRL MA-9).

The Contractor **shall** conduct a life test, within representative operational environment, to at least 2x expected life for all repetitive motion devices, with a goal of completing 1x expected life by CDR.

3.5 Workmanship

3.5.1 General

The Contractor **shall** implement a workmanship program to assure that electronic packaging technologies, processes, and workmanship meet mission objectives for quality and reliability per the requirements of the following standards:

- NASA-STD-8739.6, Implementation Requirements for NASA Workmanship Standards, excluding sections 8.1, 9.1, and 10.1
- IPC J-STD-001, Class 3
- GSFC-STD-6001
- IPC-2225
- IPC-6015

The Contractor **shall** comply with one of the following standards for electrical cables and harnesses:

- NASA-STD-8739.4
- IPC/WHMA-A-620x-S

3.5.2 Electrostatic Discharge Control

The Contractor **shall** implement an electrostatic discharge (ESD) control plan that conforms to the requirements of American National Standards Institute (ANSI)/ESD S20.20.

3.5.3 Printed Circuit Boards

The Contractor **shall** comply with one of the following standards for rigid printed circuit boards (PCB):

- IPC-6012 (Note: the latest revision preferred, but older revisions **may** be acceptable based on inherited designs or contractor standard practices)
- MIL-PRF-55110H
- ECSS-Q-ST-70-60

The Contractor **shall** implement a PCB procurement plan.

3.5.4 Lead-Free Control Measures

Tin-based solders and surface finishes that are less than 3% lead by weight **shall** comply with the Level 2C requirements set from GEIA-STD-0005-1 and GEIA-STD-0005-2.

3.6 Electrical, Electronic, and Electromechanical Parts

3.6.1 General

The Contractor **shall** document and implement a Parts Control Plan per Level 3 instructions of GSFC EEE-INST-002. The Contractor should evaluate the use of Level 2 where appropriate to meet the mission life and radiation requirements.

The Developer may use as is automotive or hi-rel commercial off the shelf parts that meet lifetime and reliability requirements with Government approval.

The Contractor **shall** perform a parts stress analysis and make available upon request.

The Contractor **shall** develop and implement a Counterfeit Parts Control Plan.

The Contractor **shall** develop and implement an Obsolescence Parts Plan.

3.6.2 Parts Control Board

The Contractor **shall** establish a PCB that is responsible for the planning, management, and coordination of the selection, application, and procurement requirements of electrical, electronic, and electromechanical (EEE) parts.

3.6.3 Re-use of Electrical, Electronic, and Electromechanical Parts

The Contractor **shall** require approval of the PCB to re-use EEE parts that have been installed.

3.6.4 Master Electrical, Electronic, and Electromechanical Parts List

The Contractor **shall** develop and deliver a Master EEE Parts List and maintain it for the duration of the project (CDRL MA-7).

3.7 Materials and Processes

3.7.1 Materials and Processes Selection, Control, and Implementation Plan

The Contractor **shall** implement a Materials and Processes (M&P) Selection, Control, and Implementation Plan in accordance with NASA-STD-6016, Standard Materials and Processes Requirements for Spacecraft, or its own internal standard which includes an Additive Manufacturing Control Plan and Part Production Plan for the design and manufacture of additively manufactured parts, in accordance with NASA-STD-6030, Additive Manufacturing Requirements for Spaceflight Systems.

3.7.2 Materials Identification and Usage List

The Contractor **shall** prepare a Materials Identification and Usage List and maintain it for the duration of the project (CDRL MA-8).

3.7.3 Materials and Processes Compliance

The use of M&P that do not comply with the requirements of NASA-STD-6016 may be acceptable in the actual hardware applications. Materials Usage Agreements **shall** be coordinated for all M&P that are technically acceptable but do not meet the requirements of NASA-STD-6016, as implemented by the approved M&P Selection, Control, and Implementation Plan.

3.8 Metrology and Calibration

3.8.1 Metrology and Calibration Program

For measurement and test equipment that has documented requirements for metrological accuracy and traceability, the Contractor **shall** comply with NASA-STD-8739.12 or verify equipment against calibrated Instruments or intrinsic standards, using a documented procedure.

Note: The Contractor **may** verify torque wrenches against a calibrated torque tester prior to use.

3.8.2 Use of Calibrated and Non-Calibrated Instruments

The Contractor **shall** record the measurements that require accuracy in applicable project build documents (e.g., work order authorizations, job orders, task sheets, or test plans), including the article of calibrated equipment used to take the measurement and its calibration end date.

When verification is chosen instead of calibration, the Contractor **shall** perform verification within a timeframe that has been demonstrated to provide appropriate levels of reliability, in the same facility, and under the same conditions that will be encountered during the process.

3.9 GIDEP Alerts and Problem Advisories

3.9.1 Government-Industry Data Exchange Program

The Contractor **shall** participate in Government-Industry Data Exchange Program (GIDEP) per the GIDEP Operations Manual (Note: this document is available through http://www.gidep.org) and support disposition and mitigation steps.

3.9.2 Alert Disposition

The Contractor **shall** review the following, hereafter referred to collectively as Alerts, for effects on EEE parts, materials, equipment, and software used in NASA products:

- GIDEP Alerts
- GIDEP SAFE-ALERTS
- GIDEP Problem Advisories
- GIDEP Agency Action Notices
- NASA Advisories

When the Contractor identifies an item in their design, inventory, or assembly that is documented in an Alert, the Contractor **shall** disposition the item and Alert through the MRB as a major nonconformance.

3.9.3 GIDEP Reporting

The Contractor **shall** prepare and submit failure experience data and safety issue reports per the requirements of GIDEP Operations Manual whenever failed or nonconforming items are discovered that are available to other buyers.

3.9.4 Review Reporting

The Contractor **shall** report the status of NASA products that are affected by Alerts (or by significant EEE parts, materials, software, and safety problems) and the actions taken to eliminate or mitigate negative effects at monthly status reviews, PCB meetings, program milestone reviews, and readiness reviews.

4 FABRICATION AND ASSEMBLY

The Contractor **shall** provide all necessary personnel, facilities, services, materials, tooling, and test equipment to fabricate and assemble the Instrument in accordance with its design specifications and Contract requirements. "Fabrication" in this sense also includes procured items.

A Manufacturing (Fabrication and Assembly) Flow Plan, or equivalent, **shall** be created and maintained and updates included as a part of reviews listed in Table 1-1, Top-Level Instrument Reviews.

4.1 Spares

The Contractor **shall** develop, deliver, and implement an approved spare parts program plan for critical flight spares and for critical GSE spares, including spares replenishment (CDRL SE-14).

Once the plan is approved by the Government, the Contractor **shall** procure sufficient spare parts to account for typical attrition during the manufacturing process due to handling, random failure, programming, etc. The Instrument spare parts **shall** be qualified, tested, stored, and calibrated to the same standards as the flight Instrument articles consistent with their spared level of assembly.

4.2 Software Development

4.2.1 Software Development Plan

The Contractor **shall** develop and control the Instrument flight software in accordance with the Instrument Software Development Plan (SDP) (CDRL SW-1). The SDP **shall** apply to software developed for this Contract, whether developed by the Contractor or by a third party (i.e., commercial-off-the-shelf, Government-off-the-shelf, modified-off-the-shelf, Subcontractor) and whether developed for flight use or for GSE/simulators.

4.2.2 Software Detailed Design [DEV]

The Contractor **shall** document and maintain Instrument GSE/STE software detailed design (CDRL SW-2).

The contractor **shall** develop a Software Version Description Document describing the deliverable code and the associated operating procedures (CDRL SW-3).

4.2.3 Software Maintenance

The Contractor **shall** develop and maintain the Instrument flight software, simulators, and documentation, along with the environments, emulators, and test software necessary to develop and verify these systems.

Any Contractor developed software development and/or validation tools (test bed) used by the Contractor to qualify and maintain the Instrument flight software **shall** be maintained by the Contractor.

The Contractor **shall** maintain any special hardware (non-commercial such as command and data handling (C&DH) components and power supplies).

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4.3 Hardware

The Contractor **shall** deliver hardware as defined in the Contract.

4.3.1 Engineering Development Program [DEV]

The Contractor **shall** conduct an Engineering Development Program to achieve TRL 6 (see NPR 7123.1, Appendix E) of all subsystems (example scan drive, digital processor, etc.), including hardware and software, and provide engineering test and characterization data for the Instrument CDR. The Engineering Development Program **shall** be documented in the Engineering Development Plan (CDRL SE-15).

4.3.2 Flight Instruments

The Contractor **shall** utilize a serial number to identify each Instrument. The flight units **shall** be built from formally released drawings, utilizing Contractor's approved parts, materials, processes, and procedures. The flight units **shall** be successfully tested by the Contractor in accordance with the Specification and the approved test procedures/levels/durations (CDRL IT-3). Test procedures may be tailored as necessary to achieve the equivalent of acceptance as defined in the ICD and PRD. Any proposed tailoring **shall** be documented in accordance with CDRL IT-1.

The Contractor **shall** provide the Instrument survival heaters that will be powered by the spacecraft.

4.3.2.1 Flight Models

The Contractor **shall** develop and deliver the Instruments as per the delivery schedule in the contract.

The FM effort **shall** include fabrication, assembly, test, maintenance, storage, delivery, post-delivery, post-launch, and on-orbit operations support per the Contract.

Qualified flight hardware that incorporate changes in design, manufacturing processing, environmental levels, or performance requirements **shall** be re-qualified.

Qualification by similarity **shall** be permitted only with the concurrence of the Government.

4.3.3 Ground Support Equipment

The Contractor **shall** develop and maintain GSE, including the STE, in accordance with the Instrument SDP (CDRL SW-1). The GSE **shall** include specialized equipment necessary to support the Instruments at all facilities during I&T, transportation, storage, satellite I&T, and launch site.

The Contractor-provided Mechanical/Electrical GSE **shall** include all items necessary for testing, deploying, stowing, and protecting the Instrument, including, but not limited to:

- STE – Interfaces to the SMBA Instrument to:
    - Provide power.
    - Monitor health and status.
    - Record and store data/files.

  - Simulate interfaces to the spacecraft (e.g., C&DH, etc.).
  - Control and monitor calibration targets.
  - Instrument commanding.
  - Provide timing.
- Hard Protective Cover for antenna apertures.
- Mechanical GSE for use during Vibration testing and Thermal Balance/TVAC testing.
- Drill template.
- Electrostatic discharge bags.
- EMI cover.
- Calibration System for Ambient Testing.
- Calibration Targets for TVAC Instrument Level Testing.
- Radio frequency blocker for EMI Testing.
- Spacecraft Emulator (delivered from the spacecraft provider).
- Flight software and Firmware Test Bed.
- Spacecraft test and operations language GSE.
- RFI generator.
- Hardware and software for reprogramming the RFI and hyperspectral configurations of SMBA.

External GSE Calibration Targets configuration **shall** be capable of accommodating Government provided targets in place of Contractor targets. Further information will be available in the Government Furnished Property List.

The Contractor **shall** design, develop, and deliver shipping and storage containers and any shipping environment-related monitoring equipment and contamination witness plates necessary for the FMs as needed based on the IMS and the Contractor's PDS Plan (CDRL IT-4) [DEV].

The Contractor **shall** provide three (3) additional sets of thermal control GSE (as needed to maintain Instrument temperature during powered operations in ambient conditions) for exclusive use at the spacecraft integration facility(ies).

4.3.4 Operations Simulator

The SMBA Operations Simulator (OpSim) is used by the NEON Series-1 team to conduct early spacecraft-to-Instrument interface checkouts; prepare the launch readiness activities; verify satellite operational loads, flight software patches, maneuver planning, ground system development, Verification and Validation (V&V) and maintenance of operational procedures and testing, flight product validation, test/operations procedure development and validation; and assist in anomaly resolution and operator training.

The Contractor **shall** develop, test, and deliver three (3) OpSim [DEV] that provides the following:

- Consists of flight-like interfaces and flight-like software.
- Provides realistic sensor operation and responses, including mission data.
- Provides capability to inject real-time or scripted anomalies.

The SMBA OpSim functionality **shall** consist of the following:

- Executes the flight-like software.
- Runs real-time and advanced time.
- Ability to suspend and resume scenario-based testing.
- Supports fault injection.
- Simulated Solid State Recorder – same operations as user interface.
- Load software and data loads to spacecraft.
- Reprogram RFI and hyperspectral configurations.

The SMBA OpSim will be used for the following:

- Testing of and production of data products.
- Used by operations to verify software and data loads to the vehicle and support operations maintenance.
- Operator Training.
- Recommended Operating Procedure (ROP)/Procedure (PROC) development.
- Interface checkout (e.g. Ethernet, 1pps, timing, etc.)
    - Commanding
    - Data
    - Telemetry

In order to support the development of scripts for use at the satellite level the OpSim **shall** be delivered prior to the Instrument PER [DEV].

The Contractor **shall** design, deliver, test, and maintain the Instrument software through the period of the contract.

5 TEST PROGRAM

The Contractor **shall** prepare and conduct a test program that meets protoflight and acceptance levels using GSFC-STD-7000, General Environmental Verification Standard (GEVS), as a guideline and as specified in the SMBA PRD. The test program **shall** verify that all requirements are satisfied. The hardware test program **shall** cover all phases of the development and manufacture in accordance with the requirements identified in this SOW and the PRD. The Contractor **shall** perform end-to-end tests in TVAC, unless otherwise approved by the Government. The Contractor **shall** provide a plan that includes any recommended tailoring for the Instrument environmental tests for Government approval in accordance with CDRL IT-1.

FM1 **shall** be tested to protoflight levels as per Section 7 of the PRD, and subsequent units **shall** be tested to acceptance levels.

The Government **shall** be invited to attend all Qualification TRRs and Formal Qualification Testing.

The Contractor **shall** provide access to facilities and data to support independent review of test data by the Government team during Instrument performance and environmental tests. Tools utilized by the Contractor to verify Instrument performance **shall** be made available to the Government.

Prior to mating with other hardware, electrical harnessing **shall** be tested to verify the wire routing, isolation, impedance, and overall workmanship.

Cabling electrically and physically equivalent to flight cables **shall** be used for all EMC testing. A Government agreed-to design, constructed with expected flight wire groupings, twisting, shielding, and shield terminations.

CPTs **shall** be conducted at the Instrument and satellite levels of assembly to demonstrate that the hardware and software meet their performance requirements within allowable tolerances. The initial CPT serves as a technical baseline against which the results of all later CPTs **shall** be compared.

CPTs **shall** demonstrate:

- With the application of known stimuli, the test article will produce the expected responses and outputs within allowable tolerances.
- The operation of redundant circuitry.
- Satisfactory performance in selected operational modes.
- When compared to the initial baseline CPT, performance of the test article has been unaffected by the environmental testing program.

When environmental testing is performed, CPTs **shall** be conducted during the hot and cold extremes of the test for both maximum and minimum input voltages.

After completion of all subassembly integration into Instruments, and after integration of the Instrument onto the satellite, a CPT will be executed in the configuration expected for

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environmental testing. Testing **shall** include operational performance evaluation at minimum and maximum flight voltages.

Limited Performance Tests (LPT) **shall** be conducted at the Instrument and satellite levels of assembly where CPTs are not warranted or not practicable to demonstrate that the selected hardware and software performance has not been degraded by environmental testing/exposure. LPTs are a subset of the overall CPT test sequence.

The LPT is a subset of tests conducted during the CPT. The Contractor **shall** describe the tests conducted for both CPTs and LPTs in the test plan (CDRL IT-1).

Electrical Aliveness Tests **shall** be conducted at the satellite level of assembly, where LPTs and CPTs are not warranted, to verify nominal telemetry response after power on and to demonstrate, to the extent possible by the test configuration, performance has not been degraded by environmental testing/exposure. Electrical Aliveness Tests are a specific subset of the overall CPT/LPT test sequence.

The Instrument **shall** be capable of operating in ambient conditions without interruption of testing at the satellite level. This may require thermal control GSE to maintain Instrument temperature during ambient testing. For satellite-level aliveness tests, the Instrument **shall** complete aliveness testing without constraining satellite operations and without requiring thermal control GSE.

The Contractor **shall** support Ground Compatibility Tests (GCT) to demonstrate the Mission Operations Center's ground system capability to communicate, command, and control the Instrument (up-link and down-link) via satellite mission communications links.

GCTs **shall** demonstrate:

- Simulated normal orbital mission scenarios encompassing launch, systems turn-on, housekeeping, command/control, and stabilization/pointing, including the collecting, processing, and archiving of science data.
- That the Instrument is immune to erroneous commands.
- Autonomous Safe Mode and simulated anomaly recovery operations.
- Reprogramming of on-board RFI and hyperspectral configurations.

Qualified flight hardware that incorporates changes in design, manufacturing processing, environmental levels, or performance requirements **shall** be re-qualified.

Qualification by similarity **shall** be permitted only with the concurrence of the Government.

5.1 Radio Frequency Interference

The Contractor **shall** test the SMBA Instrument’s capability to detect man-made RFI in the bands identified in PRD Table 4.1.2.

Rationale: The Contractor is expected to decide the best means of doing this (e.g., data sets, data ports, RFI generator, RFI Test Bench, etc.).

RFI testing **shall** include an end-to-end test.

The Contractor **shall** develop and deliver RFI algorithms (CDRLs OO-2, CV-9) and RFI test cases, and perform RFI detection analysis (CDRL CV-9) for what is currently seen from both terrestrial and space-based sources and from what can be reasonably projected in the future.

The algorithms utilized in the detection schemes **shall** be chosen by the Contractor and approved by the Government.

5.2 Antenna Pattern Measurements

The antenna radiation characteristics of each FM **shall** be measured and documented by the Contractor.

5.3 Instrument Test and Calibration [DEV]

5.3.1 Calibration Test Equipment

The Contractor **shall** provide one set of Calibration Test Equipment (CTE).

The CTE **shall** be designed and built to be used in a TVAC environment.

The CTE **shall** include calibration reference targets representing the on-orbit cold space view scene and on-orbit Earth viewing scene.

All CTE targets **shall** be large enough to fill the Instrument field of view when the Instrument is pointed toward the target.

The CTE cold space view calibration reference target(s) **shall**:

- Contain 4-wire platinum resistance thermometers (PRT) imbedded and located in the target sufficient to measure the physical temperature of the target and measure the vertical and horizontal temperature gradients within the target.

The CTE Earth viewing scene calibration reference target(s) **shall**:

- Produce a minimum of 11 radiometric brightness scene temperatures corresponding to target physical temperatures from 80K to 330K in 25K steps.
- Contain 4-wire PRTs imbedded and located in the target sufficient to measure the physical temperature of the target and measure the vertical and horizontal temperature gradients within the target in accordance with the PRD.

5.3.2 Notification of Tests

The Contractor **shall** notify the Government no later than five days in advance of Instrument and lower-level acceptance tests.

5.4 Verification and Validation

5.4.1 Verification and Validation Plan

The following delineates the baseline verification testing sequence for the SMBA Instrument and, while it cannot be deleted, the sequence of tests can be tailored as needed to optimize schedule or workflow considerations.

The baseline Instrument verification tests, prior to delivery of the Instrument, **shall** be the following:

- Electrical Interface Testing
  - Including harness testing
- CPT
  - Including LPTs
  - Including Mechanical Deployments
- EMI/EMC Characterization and Verification tests
  - Including LPTs between EMI/EMC test types
- LPT
- Structural and Dynamic Testing
  - Pre-Test Mechanical and Antenna Alignment Verifications
  - Vibroacoustic Testing
  - Sinusoidal Vibration Testing
  - Shock Testing
  - Post-Test Mechanical and Antenna Alignment Verifications
  - Note: LPTs should be performed between each Structural Test sequence (each axis or sequence where appropriate)
- CPT (pre-TVAC, and prior to each environmental test)
- TVAC and Thermal Balance Testing
  - Including Plateau CPT and LPT Testing
  - Including Performance Verification Testing
  - Including Contamination/Bake-out Testing
- CPT (post-TVAC, and after each environmental test)
  - Including Final Mechanical Alignment Verifications
- Mass Properties

The SMBA Instrument **shall** survive the Satellite-Level Test program. The baseline Instrument verification tests, after shipment to the satellite integrator, are defined as:

- Electrical Interface Tests
  - Including Electrical Aliveness testing
- CPT
  - Including Mechanical Deployments
- EMI/EMC Characterization and Verification tests
- LPT
- Structural and Dynamic Testing
  - Pre-Test Mechanical Alignment Verifications
  - Vibroacoustic Testing
  - Sinusoidal Vibration Testing

    - Shock Testing
    - Post-Test Mechanical Alignment Verifications
    - Note: LPTs should be performed between each Structural Test sequence (each axis or sequence) where appropriate.
- CPT (pre-spacecraft TVAC, and prior to each environmental test)
- TVAC and Thermal Balance Testing
    - Including Plateau CPT and LPT Testing
    - Including Performance Verification Testing
    - Including Contamination/Bake-out Testing
- CPT (post-spacecraft TVAC, and after each environmental test)
    - Including Mechanical Deployments
    - Including Final Mechanical Alignment Verifications
- Mass Properties
- GCT

The Contractor **shall** analyze and document results for those tests recommended by the Contractor that are run at satellite levels to demonstrate that the Instrument hardware and software meet their performance requirements within allowable tolerances.

6 DELIVERY

The Contractor **shall** provide a FM Instrument, with necessary GSE and personnel, to support I&T with the spacecraft. The Contractor **shall** be fully responsible and liable for the preservation, packaging, product marking, packing, and delivery of all hardware and documentation in accordance with this SOW and all applicable documents to assure safe arrival at destination.

6.1 Packaging, Handling, Storage, and Transportation

The Contractor **shall** provide Packaging, Handling, Storage, and Transportation (PHS&T) Plans and Procedures in accordance with the Instrument Post Delivery Support (PDS) Document (CDRL IT-4). The Contractor **shall** provide the personnel, facilities, and hardware necessary to prepare and pack the Instrument and its GSE for shipment and **shall** be responsible for the transportation and shipment of the material to the designated spacecraft facility, or other required location. The Contractor **shall** prepare, pack, and ship all Instrument subsystems, systems, and the simulators between the places of manufacture, I&T, storage, and delivery in accordance with Contractor best practices. The Contractor **shall** prepare, pack, and ship all related calibration, mechanical, and electrical ground support equipment (EGSE) required to support any Instrument subsystems, systems, and simulators during each phase of test, integration, and launch preparation.

PHS&T Plans (included in CDRL IT-4) **shall** include one year of storage at the Contractor's facility and long term (12 years) storage plans, and necessary storage related testing, cleaning, and maintenance for FM1 and additional FMs as required. This is for planning purposes only. See contract clause (**TBD**).

Instrument EGSE **shall** make accessible Instrument-relevant test data, such as chamber telemetry, analog test point, accelerometer, and thermocouple data.

The electrical interface between the spacecraft bus and the Instrument **shall** be checked out prior to Instrument delivery to the spacecraft using an electrical interface simulator provided by the spacecraft contractor.

6.2 Delivery, Checkout, and Acceptance

The Contractor **shall** be responsible for unpacking, inspecting, and post-shipment testing of all shipped Instrument related hardware and software prior to hand-over to the spacecraft contractor for integration. The spacecraft contractor will provide facilities, security, storage, standard cleaning supplies, and standard test equipment (multi-meter, probes, grounding, ionized air, etc.). The Contractor **shall** provide a bench checkout test report including comparison to the identical pre-ship test performed at the Contractor's facility. The Contractor **shall** compare the Instrument post-shipment functional test results with the pre-shipment functional test results and provide comparative analyses and results (CDRL IT-2).

Government Acceptance will occur at the Event/Milestone specified in Contract Section E, Clause GSFC 52.246-93 ACCEPTANCE -- LOCATION(S). The Contractor **shall** be responsible for the Instrument handling and operation through Government Acceptance. The Contractor **shall** be responsible for the Instrument handling and operation from delivery until

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lifted "on-hook" by the spacecraft contractor for mechanical integration with the spacecraft. The spacecraft contractor will be responsible for maintaining and controlling a safe environment for all Instrument hardware and personnel from the time of arrival at the spacecraft contractor's facility until delivered to the launch site, or ready to ship for return to the Instrument Contractor, as directed by the Government.

6.2.1 Documentation Delivered with Hardware

Handling instructions/procedures and accompanying documentation may be inside the outer shipment package but **shall** be outside the sealed inner package and be accessible upon opening of the outer box, without having to remove the inner box.

6.2.2 Photographic Documentation

The Contractor **shall** produce and archive closeout photographic documentation of all assemblies during the manufacturing process and of the final integrated configuration "as flown." (Include the following in meta data: name, number, date, assembly/sub-assembly, etc.)

The Contractor **shall** provide a small number of photos each month for public release.

6.2.3 Test Data Files

The Contractor **shall** provide interim test data files taken during the Instrument testing with appropriate logs for discussion of test results with the Government. The Government will provide access to an approved electronic document repository (e.g., the external SharePoint).

6.3 Sensor Database Support

6.3.1 Sensor Command and Telemetry Database Support

The Contractor **shall** collaborate with the spacecraft contractor to develop and deliver the Instrument Command and Telemetry Database (CTDB) (CDRL OO-4). The Contractor **shall** place the CTDB under formal CM.

6.4 Instrument Post-Delivery Support [DEV, TO]

The Contractor **shall** support the mission during satellite I&T, training, mission readiness support, launch site support, and operations through commissioning. CLIN applicability is specified in Contract Section C, Clause GSFC 52.211-91 SCOPE OF WORK, including what portions are covered under CLIN 0003.

The post-delivery period (from delivery of the SMBA through on-orbit support) is divided into the following sub-periods for the purposes of explicitly defining Contractor support:

- Post Delivery Support Planning (PDSP) – Activities prior to Instrument shipment.
- Post Shipment Support (PSS) – The period from successful spacecraft integration to Launch.
- Post Launch and Commissioning Support – The period beyond Commissioning or Launch + 6 months, whichever is later, through the duration of the Contract.
- Calibration/Validation Support – The period during Commissioning and in Sustaining Engineering up to Launch + 1 year.
- Sustaining Engineering – The period beyond Commissioning.

The Contractor **shall** provide satellite I&T, mission operations readiness support and training, launch site support, and support through operations hand-over in accordance with CDRL OO-1 and CDRL IT-4. The expected schedules are 24x7 during TVAC and 16x5 for other testing.

The Instrument Contractor **shall** provide the following inputs to the spacecraft contractor for Instrument GSE and flight hardware testing:

- Applicable Instrument procedures (CDRL IT-3) or inputs to spacecraft procedures that are necessary to run the GSE or flight hardware.
- The inputs to spacecraft procedures capturing all activities, equipment, personnel, and support required for the Contractor's Instrument to be ready for flight after being delivered to the spacecraft contractor **shall** be documented through a PDS document in accordance with CDRL IT-4.
- Safe-to-Mate tests during satellite I&T for FMs and GSE.
- Applicable Instrument scripts/software or inputs to spacecraft scripts/software that are necessary to run the GSE or flight hardware (CDRL IT-4).

6.4.1 General Roles and Responsibilities

Prior to launch, the Contractor **shall** be the sole responsible party for Instrument mechanical support, including, but not limited to, red and green tag removal and installation, cleaning, inspection, etc. The Contractor **shall** be responsible for the operation and physical handling of the Instrument until the spacecraft contractor accepts it for mechanical integration with the spacecraft, and whenever the Instrument is de-integrated from the spacecraft. The Contractor **shall** be responsible for operating and maintaining their GSE at the spacecraft contractor and launch site facilities. The Contractor **shall** be responsible for shipping the Instrument GSE from the launch site payload processing facility in a timely manner after launch. The Contractor **shall** be responsible for monitoring and reporting the Instrument health and safety at all times the Instrument is powered. The Contractor **shall** have personnel on-site during powered operations.

6.4.2 System Integration and Test Support

6.4.2.1 Instrument Contractor Personnel and Facility Support

The Contractor **shall** provide personnel necessary to support I&T activities at both the Contractor's facility and the spacecraft integration facility, including: performing Instrument bench checkout; providing technical support during integration of the Instrument to the spacecraft; performing all required repairs and re-tests; supporting limited and comprehensive performance testing throughout satellite qualification; performing thermal, mechanical, and electrical close-out; cleaning and monitoring contamination as needed; and supporting electrical system checkout following integration.

To prepare for the activities listed above, the Contractor **shall** support periodic virtual and in person face-to-face technical interchange meetings (TIM) with NASA and the spacecraft contractor. These TIMs will be used to define satellite LPTs, satellite CPTs, satellite environmental tests, mission readiness tests, mission simulations, and miscellaneous test details.

The Contractor **shall** provide flight qualified Instruments, with necessary GSE and personnel, to support satellite I&T. Satellite I&T will encompass mechanical integration (including alignment), electrical integration, LPT and CPT at ambient, satellite environmental qualification, and post-environmental functional and performance testing.

6.4.2.2 Instrument Performance Evaluation

The Contractor **shall** support Instrument performance evaluation (e.g., data reduction and trending) during all phases of satellite I&T, including test support, anomaly resolution support, and Instrument data archival at the Contractor's facility, with electronic access provided to the Government, as required. Data resulting from these evaluations and tools utilized by the Contractor **shall** be made available upon request by the Government.

6.4.2.3 Instrument Data Products Support

The Contractor **shall** provide the necessary Instrument operating and test procedures (CDRL IT-3), parameters, screen displays, databases (CDRL OO-4), red and yellow limits definition (CDRL OO-4), and Instrument constraints, restrictions, and warnings/alerts (Inputs to Instrument-to-Spacecraft ICD, CDRL SE-11). The Contractor **shall** provide support for the conversion and validation of these data. This support **shall** include as a minimum a review of the converted documentation (page display printouts, procedure printouts) as well as validation in a satellite test environment. Additionally, the Contractor **shall** provide Instrument safety information and may be required to support periodic safety meetings. Instrument Data Sets for use in Ground Segment testing **shall** be provided in accordance with CDRL SW-5.

The Instrument Contractor **shall** accommodate remote network access from the spacecraft contractor facility to the Instrument Contractor facility in order to receive test data efficiently.

6.4.2.4 Interface Testing

The Contractor **shall** support interface testing between their Instrument, or a reasonable facsimile (e.g. OpSim), and the spacecraft C&DH system, or a reasonable facsimile (e.g., Spacecraft-Instrument Interface Simulator (SIIS)). This testing may include use of simulators, Flight Units, or other facsimiles. The fidelity of the test units **shall** be sufficient to validate all electrical interfaces between the spacecraft and the Instrument: power bus; data bus for command, housekeeping telemetry, and high-rate science data; timing bus; and discrete analog and digital services in support of these activities. The Contractor **shall** maintain, operate, and ship Instrument GSE hardware and software to the interface testing facilities.

The Instrument Contractor **shall** support the spacecraft activity to validate the performance of the Instrument flight software by supporting satellite I&T, mission readiness, etc., during a Government approved Day-In-The-Life test which will be conducted at the spacecraft simulator provider facility.

As part of the above defined interface testing, the Contractor **shall** plan and implement support of spacecraft to payload interface electronics development risk mitigation testing. Technical support **shall** be provided by the Instrument Contractor at the spacecraft contractor facility.

As part of the above defined interface testing, the Contractor (in conjunction with the spacecraft contractor) **shall** plan and implement the testing of Instrument-spacecraft interfaces with the SIIS provided by the spacecraft contractor. Technical support will be provided by the spacecraft contractor at the Instrument simulator provider facility and any training necessary to facilitate such support by the spacecraft contractor will be provided by the Instrument simulator provider.

6.4.2.5 Bench Checkout

The Contractor **shall** provide technical support for the Instrument at the spacecraft facility. The goal of the Bench Acceptance Test (BAT) at the spacecraft facility is to certify that the Instrument has survived shipment to the spacecraft facility by replicating the pre-ship limited functional test. The Contractor **shall** define a limited functional test for the Instrument, using the Instrument GSE/STE, for use after delivery to the spacecraft contractor's facility. The Contractor **shall** be responsible for performing this test prior to integrating the Instrument to spacecraft. The Instrument EGSE **shall** be used to perform the stand-alone BAT when the Instrument arrives at the spacecraft contractor.

6.4.3 SMBA OpSim Support

The Contractor **shall** provide PDS for the OpSim, including the following:

- On-site support with initial delivery to assist in installation (five working days).
- Support installation of the OpSim software.
- Support installation of corresponding simulator Telemetry and Command and Simulator Control databases.
- Assist in basic checkout of simulator interfaces and functions.
- Phone/E-mail support afterward and with subsequent updates.
- Satellite compatibility testing.

The Instrument OpSim **shall** be updated concurrently with the Instrument flight software.

6.4.4 Launch Site Support

The Contractor **shall** provide launch site support at the payload processing facility as necessary to ensure a successful and timely launch, including the following:

- Limited satellite, including Instrument, functional testing prior to integration to the launch vehicle.
- Final thermal, electrical, and mechanical closeout of the Instrument.
- Final red tag/green tag removal and installation.
- Final Instrument inspection and cleaning.
- Anomaly support.

6.4.5 Mission System Integration Support

The Contractor **shall** support planning and execution of mission systems integration activities through commissioning, including training and mission readiness support as described below.

The Contractor **shall** provide training for the Mission Elements Operations Team.

6.4.5.1 Integrated Mission Timeline

The Contractor **shall** support development of Integrated Mission Timeline (IMT), including:

- Support for regular IMT Working Group.
- Review IMT prior to major stakeholder reviews and attend reviews.

6.4.5.2 Rehearsal Anomaly Definition

The Contractor **shall** support Rehearsal Anomaly Definition, including:

- Define anomaly scenarios to inject into rehearsals.
- Assist in scripting anomalies for the simulator.

6.4.5.3 Satellite Operations Exercises

The Contractor **shall** participate in Satellite Operations Exercises (SOE), including:

- Participate in planning of Instrument-focused SOEs.
- Review and approve Test Plans, Procedures, and Configurations.
- Support TRR.
- Monitor execution of Instrument-focused SOEs at spacecraft facility and the satellite operations facility.

6.4.5.4 Compatibility Tests

The Contractor **shall** participate in satellite-level compatibility tests, including:

- Participate in planning of Instrument-focused compatibility.
- Develop, review, and approve Test Plans, Procedures, and Configurations.
- Support Readiness Review.
- Participate in Early Engineering Opportunity, dry-run, and run-for-record execution of tests.

6.4.5.5 Mission Rehearsals

The Contractor **shall** participate in Mission Rehearsals, including:

- Three (3) Instrument-focused rehearsals and dress rehearsals.
- Three (3) Support Operations Readiness Exercises as needed to prepare for and/or supplement Mission Rehearsals.

6.4.5.6 Operations Product Validation

The Contractor **shall** support Operations Product Validation, including:

- Review PROCs for consistency with ROPs.
- Review other ops products such as pages and tasks for accuracy and contingency procedures.
- Witness execution of final PROCs to assess readiness for flight.

6.4.5.7 Recommended Operating Procedures

The Contractor **shall** develop Instrument ROPs (CDRL OO-5) and:

- Recommend ROP changes as needed based on lessons learned.
- Review any Mission Operations Team ROP changes based on operational lessons learned.
- Update ROPs as needed.

6.4.5.8 Operational Readiness Test

The Contractor **shall** support an Operational Readiness Test (ORT) that includes the review of Instrument-specific information in the ORT Plan for accuracy and completeness.

6.4.6 Commissioning Support

The Contractor **shall** support Satellite Launch, Early Orbit, and Anomaly checkout, including providing necessary engineering support at the operations facility for on-orbit activation and checkout of the Instrument. The commissioning period is nominally 90 days. The Contractor **shall** provide, at a minimum:

- Instrument support during bus commissioning at the operations facility (monitoring Instrument survival telemetry, spacecraft outgassing, and other Instrument concerns as needed).
- Instrument participation in real-time go/no-go decision-making during Instrument activation at the operations facility.
- Support during buy-off.
- Anomaly resolution support.
- Support during Instrument calibration and validation.
- Verification of SMBA health, functionality, and performance. Evaluations to include trending and comparisons to requirements and expectations established during ground test and characterizations. The results of all evaluations **shall** be documented in the Commissioning Report (CDRL OO-3).

6.4.7 Post-Commissioning Sustaining Engineering

The Contractor **shall** support the mission by providing:

- Anomaly resolution support through the period of performance of the Contract.
- Software maintenance through the period of performance of the Contract (includes Flight, Simulator, and GSE software).
- RFI detection and hyperspectral reprogramming.

Appendix A ABBREVIATIONS AND ACRONYMS

Abbreviation/Acronym	Definition
ARB	Anomaly Review Board
ANSI	American National Standards Institute
ATMS	Advanced Technology Microwave Sounder
BAT	Bench Acceptance Test
BCR	Baseline Change Request
C&DH	Command and Data Handling
CCB	Configuration Control Board
CCR	Configuration Change Request
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CLIN	Contract Line Item Number
CM	Configuration Management
CPT	Comprehensive Performance Test
CTB	Consent to Break
CTDB	Command and Telemetry Database
CTE	Calibration Test Equipment
CTP	Consent to Proceed
CV	Calibration/Validation
DB	Direct Broadcast
DEV	Development
DID	Data Item Description
DOC	Department of Commerce
EEE	Electrical, Electronic, and Electromechanical
EGSE	Electrical Ground Support Equipment
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EPR	Engineering Peer Review
ERB	Engineering Review Board
ESD	Electrostatic Discharge
EVM(S)	Earned Value Management (System)
FAR	Federal Acquisition Requirements
FM	Flight Model
FOD	Foreign Object Damage/Debris
FPGA	Field Programmable Gate Array
FRB	Failure Review Board
GCT	Ground Compatibility Test
GEVS	GSFC Environmental Verification Specifications

GIDEP	Government-Industry Data Exchange Program
GPR	Goddard Procedural Requirement
GR	Giver/Receiver
GSE	Ground Support Equipment
GSFC	Goddard Space Flight Center
G-R	Giver/Receiver
Hz	hertz
I&T	Integration and Test
IBR	Integrated Baseline Review
ICD	Interface Control Document
IMS	Integrated Master Schedule
IMT	Integrated Mission Timeline
IPMDAR	Integrated Program Management Data and Analysis Report
ISAR	Instrument Safety Assessment Report
IT	Integration and Test
JPSS	Joint Polar Satellite System
K	kelvin
LEO	Low Earth Orbit
LPD	Low Earth Orbit Programs Division
LPT	Limited Performance Test
M&P	Materials and Processes
MA	Mission Assurance
MICD	Mechanical Interface Control Document
MRB	Material Review Board
MRR	Manufacturing Readiness Review
MW	LEO Microwave
MWS	Microwave Sounder
N/A	Not Applicable
NASA	National Aeronautics and Space Administration
NEdT	Noise Equivalent Differential Temperature
NEON	Near Earth Orbit Network
NESDIS	National Environmental Satellite, Data, and Information Service (NOAA)
NIST	National Institute for Standards and Technology
NOAA	National Oceanic and Atmospheric Administration
NPR	NASA Procedural Requirement
NS1	NEON Series-1
NSOSA	NOAA Satellite Observing System Architecture
OO	On Orbit
OpSim	Operations Simulator

ORT	Operational Readiness Test
PCB	Parts Control Board
PCB	Printed Circuit Board
PDR	Preliminary Design Review
PDS	Post Delivery Support
PDSP	Post Delivery Support Planning
PER	Pre-Environmental Review
PHS&T	Packaging, Handling, Storage, and Transportation
PM	Program Manager/Management
PMEF	Primary Mission Essential Functions
PRD	Performance Requirements Document
PROC	Procedure
PROD	Production
PRT	Platinum Resistance Thermometer
PSR	Pre-Ship Review
q	Moisture
QMS	Quality Management System
RDR	Raw Data Record
RE	Review
RFI	Radio Frequency Interference
ROP	Recommended Operating Procedure
SAE	Society of Automotive Engineers
SDP	Software Development Plan
SDR	System Definition Review
SE	Systems Engineering
SIIS	Spacecraft-Instrument Interface Simulator
SMBA	Sounder for Microwave-Based Applications
SOE	Satellite Operations Exercise
SOW	Statement of Work
SRF	Spectral Response Function
SRR	Systems Requirements Review
STE	System Test Equipment
SW	Software
T	Temperature
TBD	To Be Determined/Developed
TBP	To Be Proposed
TBR	To Be Reviewed
TBS	To Be Supplied
TDR	Test Data Review
TIM	Technical Interchange Meeting

TO	Task Order
TPM	Technical Performance Metric
TRL	Technology Readiness Level
TRR	Test Readiness Review
TVAC	Thermal Vacuum
V&V	Verification and Validation